TEXAS
The University of Texas at Austin

CS439 - Principles of Computer Systems

53215

Spring 2026

CLASS MEETS: MW 4pm-6pm, CPE 2.204, F 10:00am-noon, UTC 3.110

Instructor: Ahmed Gheith
Office: GDC 5.302
TA and office hours information announced on Canvas/Ed

Course Description

An introduction to computer systems software abstractions with an emphasis on the connection of these abstractions to underlying computer hardware. Key abstractions include threads, virtual memory, protection, and I/O. Requires writing of synchronized multithreaded programs and pieces of an operating system.

FLAG COURSES

May be counted toward the Independent Inquiry flag requirement.

How Will You Learn?

You will mostly learn by doing the programming assignments. Lectures and discussions will help you tackle the assignments and might mislead you into believing that you understand the material but it will not sink in until you finish the assignments

TEACHING MODALITY INFORMATION

Lectures and discussions will be in person (Zoom will be utilized for some office hours and in case of unforeseen emergencies).

COMMUNICATION

The course Canvas site can be found at https://utexas.instructure.com/courses/1402350. Please email me through Canvas. You are responsible for ensuring that the primary email address you have recorded with the university is the one you will check for course communications because that is the email address that Canvas uses.

ASKING FOR HELP

Start by asking in the public class discussion group: https://edstem.org/us/courses/64631/discussion/ or send a private message if it involves information you don't want to share publicly.

TEXAS
The University of Texas at Austin

DISABILITY & ACCESS (D&A)

The university is committed to creating an accessible and inclusive learning environment consistent with university policy and federal and state law. Please let me know if you experience any barriers to learning so I can work with you to ensure you have equal opportunity to participate fully in this course. If you are a student with a disability, or think you may have a disability, and need accommodations please contact Disability & Access (D&A). Please refer to the D&A website for more information: http://diversity.utexas.edu/disability/. If you are already registered with D&A, please deliver your Accommodation Letter to me as early as possible in the semester so we can discuss your approved accommodations and needs in this course.

Course Requirements and Grading

REQUIRED MATERIALS

OPERATING SYSTEMS: PRINCIPLES AND PRACTICE, 2ND EDITION, THOMAS ANDERSON, MICHAEL DARLIN, **ISBN-13:** 978-0985673529

REQUIRED DEVICES

The public CS Linux machines is all you need https://www.cs.utexas.edu/facilities/public-labs. You can use your own desktop and/or laptop for development but it is your responsibility to ensure that your submission works on the CS machines

CLASSROOM EXPECTATIONS

Class participation 10% if your grade will depend on class participation. This includes:

* Asking questions during lectures and discussions and on the discussion group
* Providing insight during such discussions
* Suggesting good quiz/test questions and assignment test cases
* Leadership in organizing and implementing the final group project

Behavior expectations

* Respect your peers
* Obtain a mutex lock before you speak during lectures and discussions
* All submitted work must be your original work. As an exception you're allowed to submit any code given to you by the instructors but you're expected to be able to answer questions about what it does and how it does it.
* You're allowed to consult with your peers and with online resources provided that:
    * You don't share code, only design and ideas.
    * You clearly cite your resources in your report.

ASSIGNMENTS

* parallelism and critical sections
* Cooperative multi-threading

* Preemptive multi-threading
* High-level synchronization primitives
* Implementing the ext2 read path
* Virtual memory
* ELF loading and system calls
* Fork, exec and other system calls
* Networking
* A simple shell
* An operating system that can run a complex program (e.g. networked doom) [group project]

LATE WORK AND MAKING UP MISSED WORK

Late submissions will be penalized at a rate of 10% (compound) per day.

GRADE COMPUTATION

Programming assignments (50%) [Almost weekly, exceptions will be announced]
Quizzes (40%) [every 2 weeks]
Class participation (10%)

GRADE BREAKS

Grade	Cutoff
A	94%
A-	90%
B+	87%
B	84%
B-	80%
C+	77%
C	74%
C-	70%
D+	67%
D	64%
D-	60%
F	<60%

Course Outline

All instructions, assignments, readings, rubrics and essential information will be on the Canvas website at utexas.instructure.com. Check Canvas regularly. **Changes** to the schedule may be made at my discretion if circumstances require. I will announce any such changes in class and will also communicate them via a Canvas announcement. It is your responsibility to note these changes when announced, and I will do my best to ensure that you are notified of changes with as much advance notice as possible.

TEXAS
The University of Texas at Austin

[Syllabus must include all major course requirements and assignments, along with the dates of exams and assignments that count for 20% or more of the class grade. Also, recall that per the General Information Catalog no exam counting for more than 30% of the final course grade may be given during the last week of class, or during no-class days/reading days preceding the final exam period.]

Week	Class Topic
1.	Overview of operating systems and concurrency
2	Cooperative multi-threading and synchronization
3	Preemptive multi-threading and synchronization
4	High level synchronization primitives
5	File systems
6	Virtual memory
7	File systems + virtual memory
8	Executable files and loading
9	Processes and system calls
10	Authentication and authorization
11	Access control lists, capabilities, etc.
12	Scheduling
13	Revisiting deadlocks, race conditions, and synchronization
14	Networking
15	Virtual machines, containers, etc

Course Policies and Disclosures

HONOR CODE

The University of Texas at Austin strives to create a dynamic and engaging community of teaching and learning where students feel intellectually challenged; build knowledge and skills; and develop critical thinking, creativity, and intellectual curiosity. As a part of this community, it is important to engage in assignments, exams, and other work for your classes with openness, integrity, and a willingness to make mistakes and learn from them. The UT Austin honor code champions these principles:

I pledge, as a member of the University of Texas community, to do my work honestly, respectfully, and through the intentional pursuit of learning and scholarship.

The honor code affirmation includes three additional principles that elaborate on the core theme:

- I pledge to be honest about what I create and to acknowledge what I use that belongs to others.

- I pledge to value the process of learning in addition to the outcome, while celebrating and learning from mistakes.
- This code encompasses all of the academic and scholarly endeavors of the university community.

The honor code is more than a set of rules, it reflects the values that are foundational to your academic community. By affirming and embracing the honor code, you are both upholding the integrity of your work and contributing to a campus culture of trust and respect.

ACADEMIC INTEGRITY EXPECTATIONS

Students who violate University rules on academic misconduct are subject to the student conduct process. A student found responsible for academic misconduct may be assigned both a status sanction and a grade impact for the course. The grade impact could range from a zero on the assignment in question up to a failing grade in the course. A status sanction can range from a written warning, probation, deferred suspension and/or dismissal from the University. To learn more about academic integrity standards, tips for avoiding a potential academic misconduct violation, and the overall conduct process, please visit the Student Conduct and Academic Integrity website at: http://deanofstudents.utexas.edu/conduct.

[It is strongly recommended that you outline any individual expectations for assignment completion- including parameters around group work, authorized resources, citation requirements, etc. in the assignment directions. Clear and detailed expectations not only reduce the likelihood of a possible violation, but they also aid the Student Conduct team in holding students accountable that fail to adhere to the assignment directions.]

CONFIDENTIALITY OF CLASS RECORDINGS

Class recordings are reserved only for students in this class for educational purposes and are protected under FERPA. The recordings should not be shared outside the class in any form. Violation of this restriction by a student could lead to Student Misconduct proceedings.

GETTING HELP WITH TECHNOLOGY

Students needing help with technology in this course should contact the ITS Service Desk

SHARING OF COURSE MATERIALS IS PROHIBITED

[Anyone concerned about the unauthorized sharing of their course materials through online sites should add the following statement to their syllabus. Including this statement regarding the posting and/or use of your course content by students establishes a proactive directive to students that is helpful to the office of Student Conduct and Academic Integrity in administering the conduct process should the material be posted in opposition to your written directive.]

No materials used in this class, including, but not limited to, lecture hand-outs, videos, assessments (quizzes, exams, papers, projects, homework assignments), in-class materials, review sheets, and additional problem sets, may be shared online or with anyone outside of the class without explicit, my written permission. Unauthorized sharing of materials may facilitate cheating. The University is aware of the sites used for sharing materials, and any materials found online that are associated with you, or any suspected unauthorized sharing of materials, will be reported to Student Conduct and Academic Integrity in the Office of the Dean of Students. These reports can result in initiation of the student conduct process and include charge(s) for academic misconduct, potentially resulting in sanctions, including a grade impact.

ARTIFICIAL INTELLIGENCE

The use of artificial intelligence tools (such as ChatGPT) for programming assignments is permitted for study and exploration purposes, provided that:

* The submitted code is written from scratch by the student
* The portions of the code helped by the use of such tools is clearly marked with block comments in the submitted code
* The student is willing and able to fully explain their answer and write it again without consulting the submitted code.

If you are considering the use of AI writing tools but are unsure if you are allowed or the extent to which they may be utilized appropriately, please ask."

For more information about AI in education, see the Center for Teaching and Learning's "5 Things to Know about ChatGPT" webpage that includes additional suggested syllabi statements for your consideration.]

RELIGIOUS HOLIDAYS

By UT Austin policy, you must notify me of your pending absence for a religious holiday as far in advance as possible of the date of observance. If you must miss a class, an examination, a work assignment, or a project in order to observe a religious holy day, you will be given an opportunity to complete the missed work within a reasonable time after the absence.

TITLE IX DISCLOSURE

Beginning January 1, 2020, Texas Education Code, Section 51.252 (formerly known as Senate Bill 212) requires all employees of Texas universities, including faculty, to report to the Title IX Office any information regarding incidents of sexual harassment, sexual assault, dating violence, or stalking that is disclosed to them. Texas law requires that all employees who witness or receive information about incidents of this type (including, but not limited to, written forms, applications, one-on-one conversations, class assignments, class discussions, or third-party reports) must report it to the Title IX Coordinator. Before talking with me, or with any faculty or staff member about a Title IX-related incident, please remember that I will be required to report this information.

Although graduate teaching and research assistants are not subject to Texas Education Code, Section 51.252, they are mandatory reporters under federal Title IX regulations and are required to report a wide range of

behaviors we refer to as sexual misconduct, including the types of misconduct covered under Texas Education Code, Section 51.252. Title IX of the Education Amendments of 1972 is a federal civil rights law that prohibits discrimination on the basis of sex – including pregnancy and parental status – in educational programs and activities. The Title IX Office has developed supportive ways and compiled campus resources to support all impacted by a Title IX matter.

If you would like to speak with a case manager, who can provide support, resources, or academic accommodations, in the Title IX Office, please email: supportandresources@austin.utexas.edu. Case managers can also provide support, resources, and accommodations for pregnant, nursing, and parenting students.

For more information about reporting options and resources, please visit: https://titleix.utexas.edu, contact the Title IX Office via email at: titleix@austin.utexas.edu, or call 512-471-0419.

CAMPUS SAFETY

The following are recommendations regarding emergency evacuation from the Office of Emergency Management, 512-232-2114:

- Students should sign up for Campus Emergency Text Alerts at the page linked above.
- Occupants of buildings on The University of Texas at Austin campus must evacuate buildings when a fire alarm is activated. Alarm activation or announcement requires exiting and assembling outside.
- Familiarize yourself with all exit doors of each classroom and building you may occupy. Remember that the nearest exit door may not be the one you used when entering the building.
- Students requiring assistance in evacuation shall inform their instructor in writing during the first week of class.
- In the event of an evacuation, follow the instruction of faculty or class instructors. Do not re-enter a building unless given instructions by the following: Austin Fire Department, The University of Texas at Austin Police Department, or Fire Prevention Services office.
- For more information, please visit the Office of Emergency Management.

University Resources

For a list of university resources that may be helpful to you as you engage with and navigate your courses and the university, see the University Resources Students Canvas page.